

DATA ANALYTICS

PYTHON ASSIGNMENT 2 :DATA STRUCTURES- LIST, DICTIONARY, SET & CONDITIONAL STATEMENTS

SUBMISSION:03.10.2026

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Data Structures - List, Dictionary, Set & Conditional Statements

List (Creation, Modification and Access):

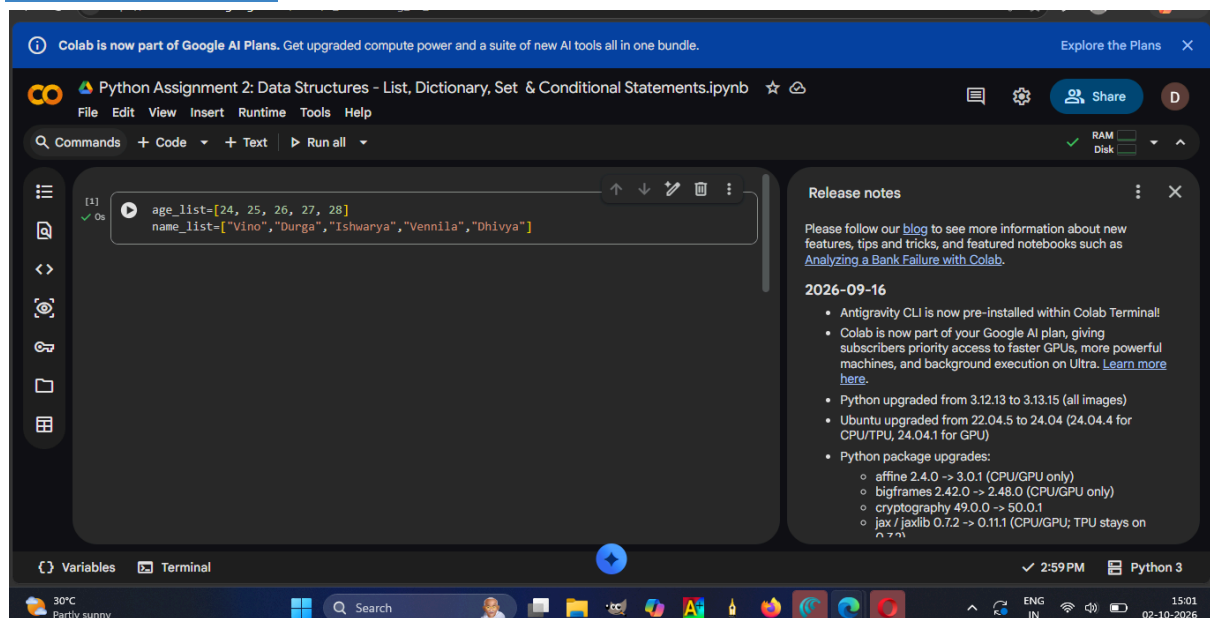
1. List Creation:

Task:

- Create a list named `age_list` with five integer elements. For eg., [24, 25, 26, 27, 28]
- Create a list named `name_list` with five string elements.

Solution :

Preview screenshot :



Explanation:

- First I had created the list with five integer elements and named the list as `age_list`.

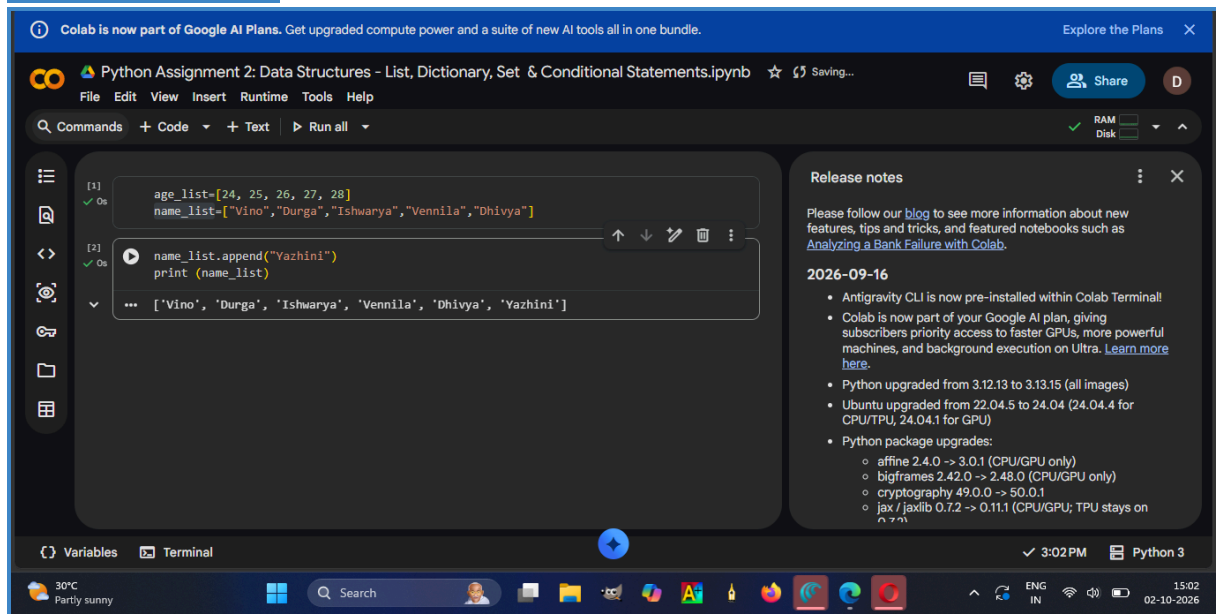
- Then created the name_list with five string elements.
- The above mentioned screenshot represents the result.

2. List Operations / Modifications:

Task 1: Append the string "Yazhini" to name_list.

Solution :

Preview screenshot :



The screenshot shows a Google Colab notebook titled "Python Assignment 2: Data Structures - List, Dictionary, Set & Conditional Statements.ipynb". The notebook has two code cells. Cell [1] contains the following code:

```
age_list=[24, 25, 26, 27, 28]
name_list=["Vino","Durga","Ishwarya","Vennila","Dhivya"]
```

 Cell [2] contains the following code:

```
name_list.append("Yazhini")
print (name_list)
```

 The output of cell [2] is a list:

```
['Vino', 'Durga', 'Ishwarya', 'Vennila', 'Dhivya', 'Yazhini']
```

 On the right side of the notebook, there is a "Release notes" panel for the date 2026-09-16, listing updates to Antigravity CLI, Colab's integration with Google AI, Python version (3.12.13 to 3.13.15), Ubuntu version (22.04.5 to 24.04), and various Python package upgrades (affine, bigframes, cryptography, jax/jaxlib).

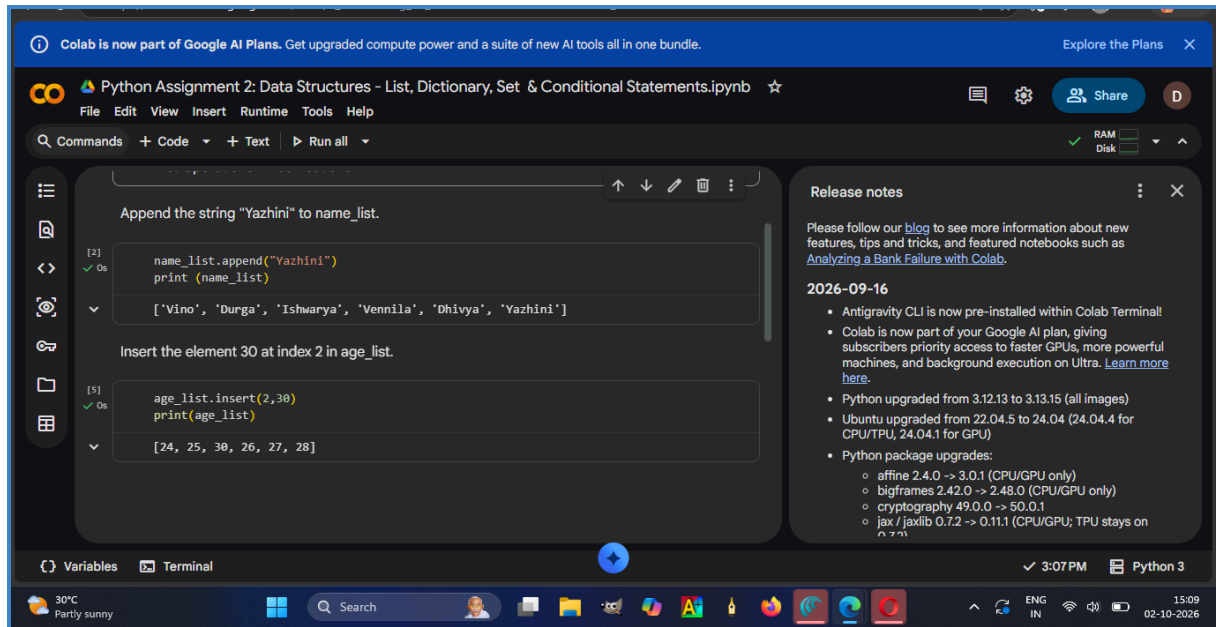
Explanation :

- I had used the append method for adding a string to name_list.
- The above mentioned screenshot represents the result.

Task 2 : Insert the element 30 at index 2 in age_list.

Solution:

Preview screenshot :



The screenshot shows a Google Colab notebook titled "Python Assignment 2: Data Structures - List, Dictionary, Set & Conditional Statements.ipynb". The notebook has two code cells. The first cell, labeled [2], contains the code to append the string "Yazhini" to a list named name_list and print it. The output shows the list ['Vino', 'Durga', 'Ishwarya', 'Vennila', 'Dhivya', 'Yazhini']. The second cell, labeled [5], contains the code to insert the element 30 at index 2 of a list named age_list and print it. The output shows the list [24, 25, 30, 26, 27, 28]. On the right side of the notebook, there is a "Release notes" panel for the date 2026-09-16, listing updates to Colab, Python, and various packages. The bottom of the screen shows a Windows taskbar with the date 02-10-2026 and time 15:09.

```
Append the string "Yazhini" to name_list.  
[2]  
✓ Os  
name_list.append("Yazhini")  
print (name_list)  
['Vino', 'Durga', 'Ishwarya', 'Vennila', 'Dhivya', 'Yazhini']  
  
Insert the element 30 at index 2 in age_list.  
[5]  
✓ Os  
age_list.insert(2,30)  
print(age_list)  
[24, 25, 30, 26, 27, 28]
```

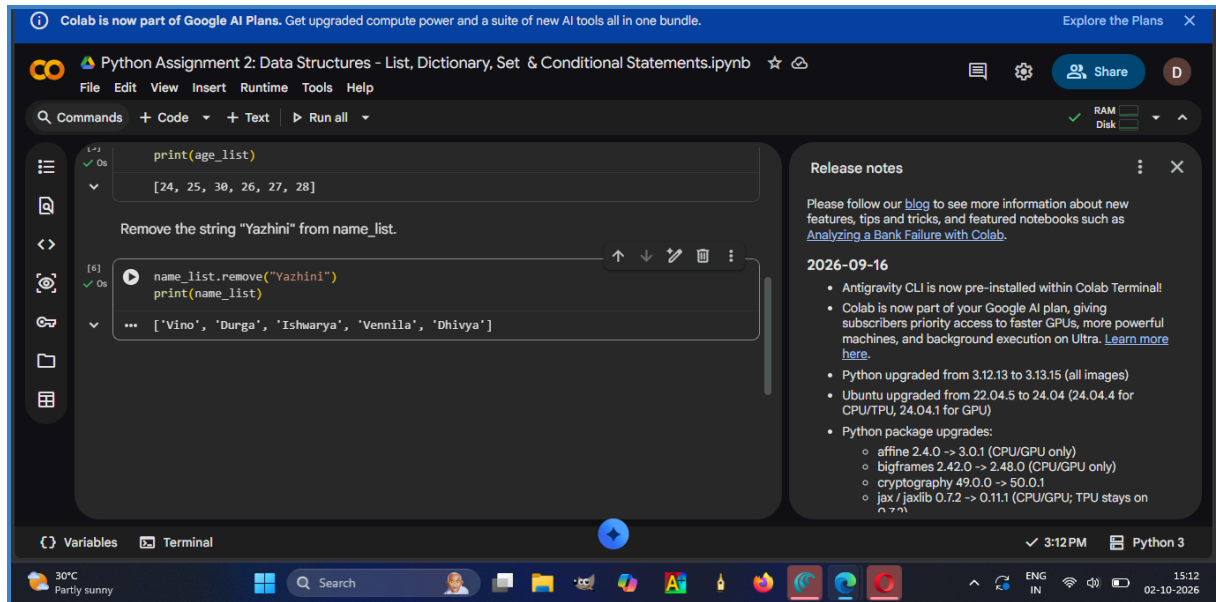
Explanation:

- I had used the insert method for inserting an element into age_list.
- The above mentioned screenshot represents the result.

Task 3: Remove the string "Yazhini" from name_list.

Solution:

Preview screenshot :



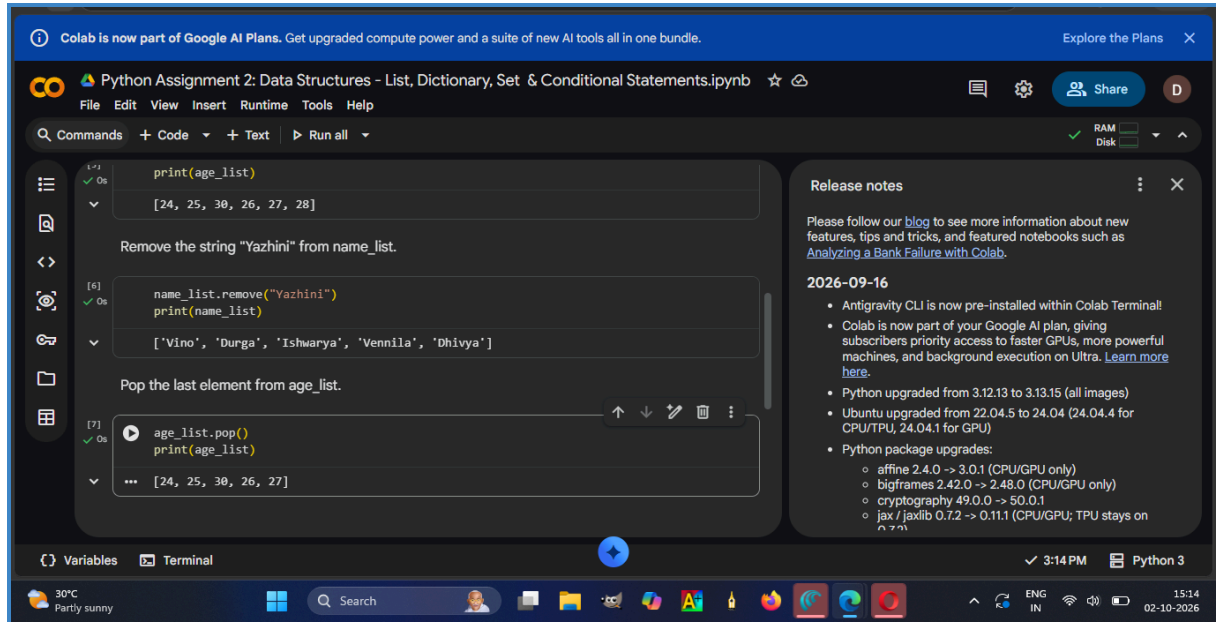
Explanation :

- I had used the insert method for inserting a string into name_list.
- The above mentioned screenshot represents the result.

Task 4: Pop the last element from age_list.

Solution:

Preview screenshot :



The screenshot shows a Google Colab notebook titled "Python Assignment 2: Data Structures - List, Dictionary, Set & Conditional Statements.ipynb". The notebook has three code cells. The first cell prints the initial age_list: [24, 25, 30, 26, 27, 28]. The second cell removes the string "Yazhini" from name_list and prints the updated list: ['Vino', 'Durga', 'Ishwarya', 'Vennila', 'Dhivya']. The third cell pops the last element from age_list and prints the updated list: [24, 25, 30, 26, 27]. A "Release notes" panel on the right side of the notebook provides information about updates to the Colab environment, including the addition of Antigravity CLI, Colab's integration with Google AI Plans, and various package upgrades for Python, Ubuntu, and other dependencies.

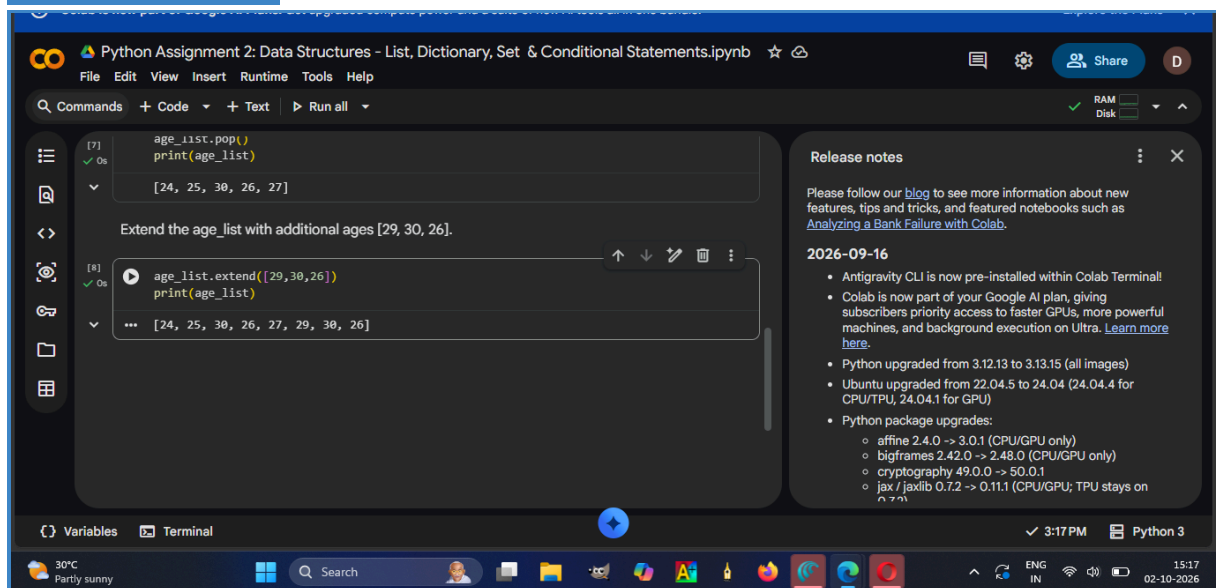
Explanation :

- I had used the pop method for deleting an element into age_list.
- The above mentioned screenshot represents the result.

Task 5 : Extend the age_list with additional ages [29, 30, 26].

Solution :

Preview screenshot :



The screenshot shows the same Google Colab notebook as before, but with a fourth code cell. This cell extends the age_list with the additional ages [29, 30, 26] using the extend() method and prints the final list: [24, 25, 30, 26, 27, 29, 30, 26]. The "Release notes" panel on the right side of the notebook remains visible, providing information about updates to the Colab environment.

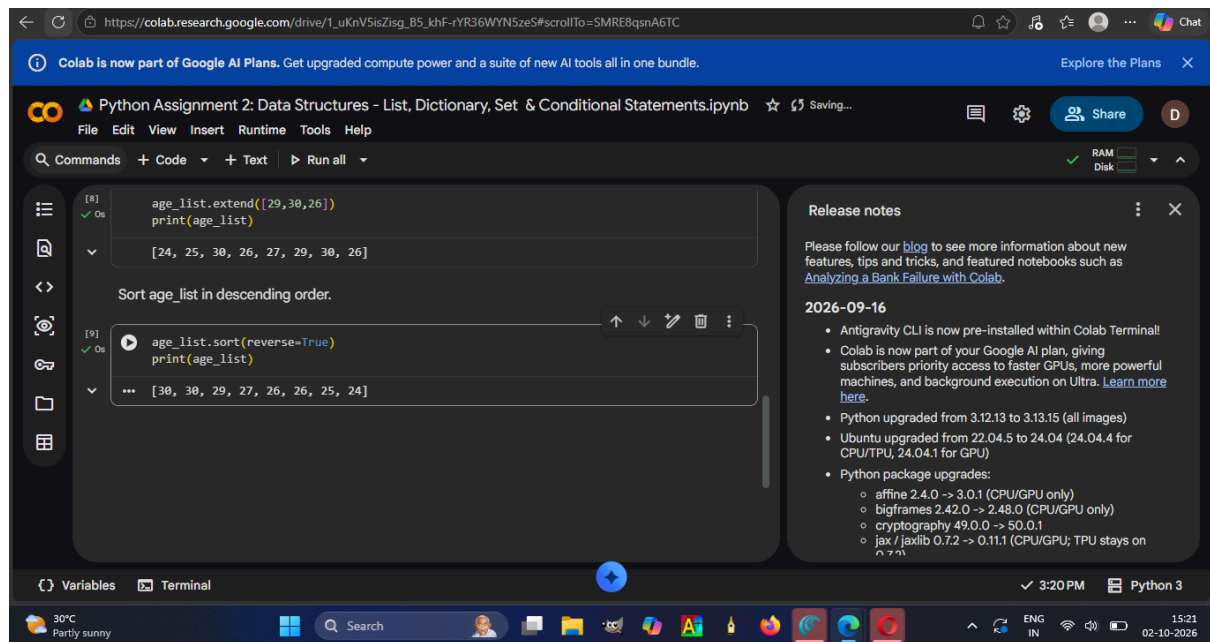
Explanation :

- I had used the extend method to add [29, 30, 26] into age_list.
- The above mentioned screenshot represents the result.

Task 6 : Sort age_list in descending order.

Solution :

Preview screenshot :



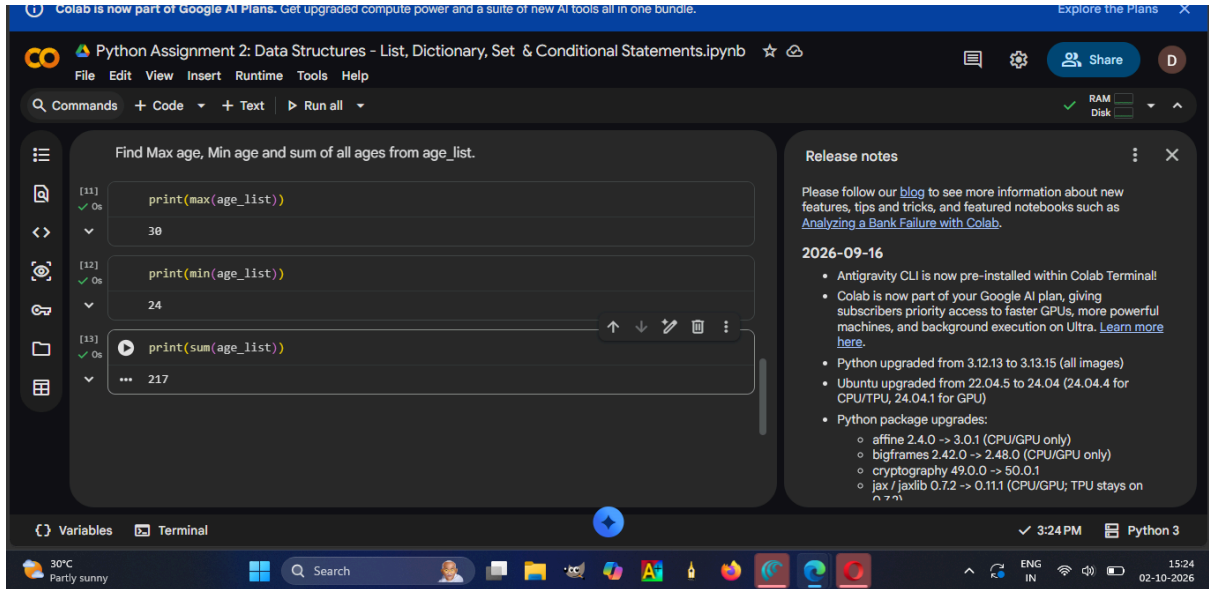
Explanation :

- I had used the sort method to sort age_list with reverse=true so it sorted the age list in descending order.
- The above mentioned screenshot represents the result.

Task 7: Find Max age, Min age and sum of all ages from age_list.

Solution :

Preview screenshot :



The screenshot shows a Google Colab notebook titled "Python Assignment 2: Data Structures - List, Dictionary, Set & Conditional Statements.ipynb". The notebook contains three code cells, each with a print statement and its output:

```
[11] print(max(age_list))
30

[12] print(min(age_list))
24

[13] print(sum(age_list))
217
```

On the right side of the notebook, there is a "Release notes" panel for the date 2026-09-16. It lists several updates, including the integration of Antigravity CLI, Colab's inclusion in Google AI plans, Python version upgrades from 3.12.13 to 3.13.15, and Ubuntu version upgrades from 22.04.5 to 24.04. It also lists Python package upgrades for affine, bigframes, cryptography, and jax/jaxlib.

The bottom of the screen shows a Windows taskbar with the date 02-10-2026 and time 15:24.

Explanation :

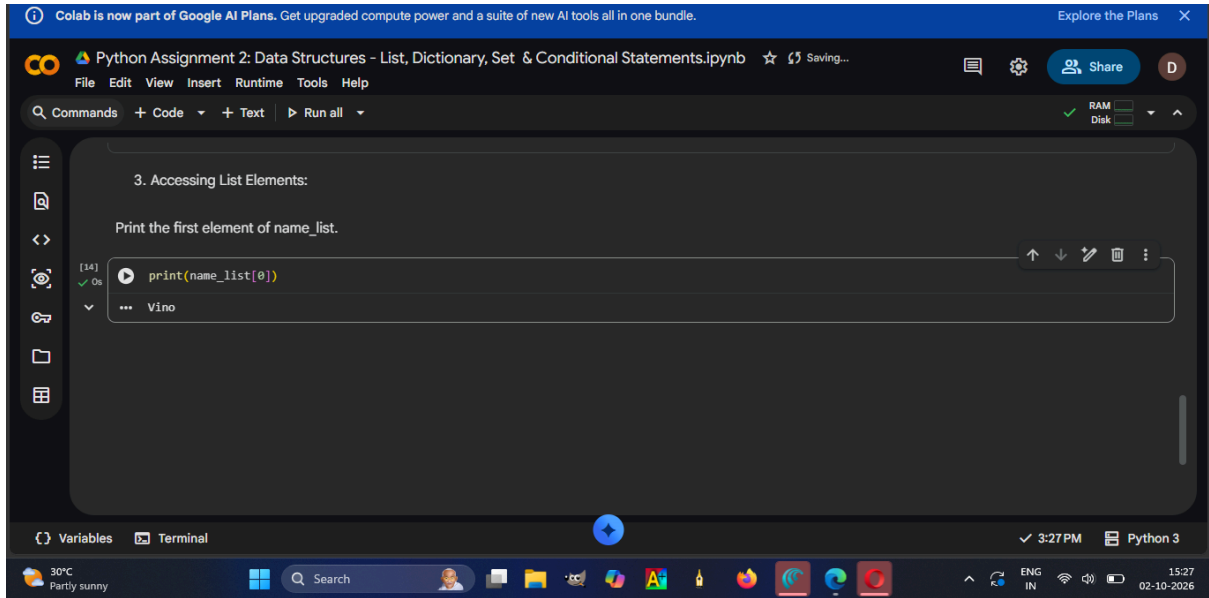
- I had used the max method to find the maximum age from the age_list. And a method to find minimum age from the age_list. And a sum method to find the sum of all ages from age_list.
- The above mentioned screenshot represents the result.

3. Accessing List Elements:

Task 1: Print the first element of name_list.

Solution :

Preview screenshot :



The screenshot shows a Google Colab notebook interface. The top bar indicates 'Colab is now part of Google AI Plans. Get upgraded compute power and a suite of new AI tools all in one bundle.' The notebook title is 'Python Assignment 2: Data Structures - List, Dictionary, Set & Conditional Statements.ipynb'. The menu bar includes File, Edit, View, Insert, Runtime, Tools, and Help. The left sidebar shows icons for file management and a search bar. The main code cell contains the text '3. Accessing List Elements:' followed by 'Print the first element of name_list.' Below this, a code input field shows the command `print(name_list[0])`. The output of the cell is displayed as 'Vino'. The bottom status bar shows 'Variables', 'Terminal', '3:27 PM', and 'Python 3'. The Windows taskbar at the very bottom shows the date '02-10-2026' and time '15:27'.

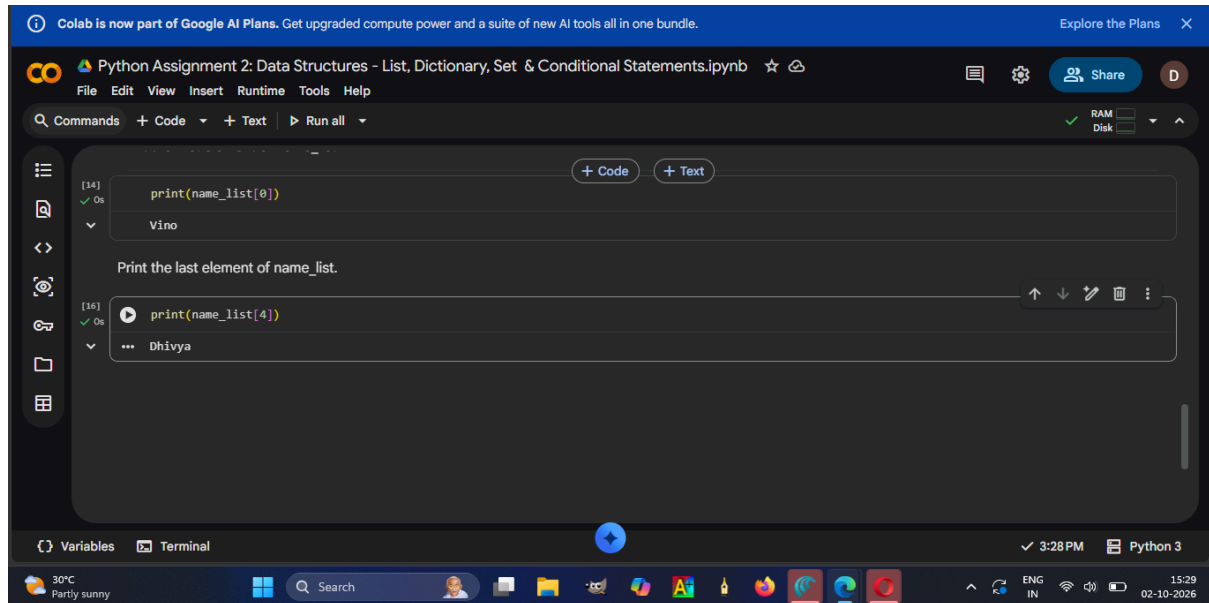
Explanation :

- I had used the indexing method for printing the first element in name_list.
- Here 0 represents the first element in the name_list.
- The above mentioned screenshot represents the result.

Task 2:Print the last element of name_list.

Solution :

Preview screenshot :



```
[14] ✓ Os
print(name_list[0])
Vino

Print the last element of name_list.

[16] ✓ Os
print(name_list[4])
... Dhivya
```

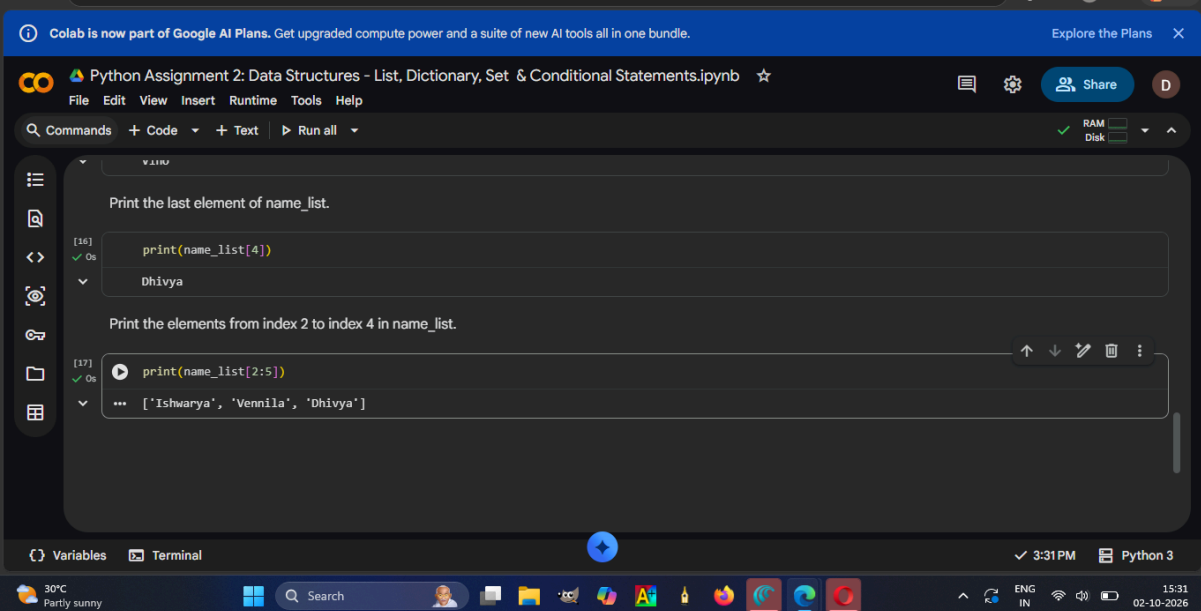
Explanation :

- I had used the indexing method for printing the last element in name_list.
- Here 4 represents the last element in the name_list.
- The above mentioned screenshot represents the result.

Task 3 :Print the elements from index 2 to index 4 in name_list.

Solution :

Preview screenshot :



The screenshot shows a Google Colab notebook titled "Python Assignment 2: Data Structures - List, Dictionary, Set & Conditional Statements.ipynb". It contains two code cells. The first cell, labeled [16], has the comment "Print the last element of name_list." and the code `print(name_list[4])`, which outputs "Dhivya". The second cell, labeled [17], has the comment "Print the elements from index 2 to index 4 in name_list." and the code `print(name_list[2:5])`, which outputs the list `['Ishwarya', 'Vennila', 'Dhivya']`. The notebook interface includes a menu bar (File, Edit, View, Insert, Runtime, Tools, Help), a toolbar with icons for running and saving, and a status bar at the bottom showing the time as 3:31 PM and the Python version as Python 3.

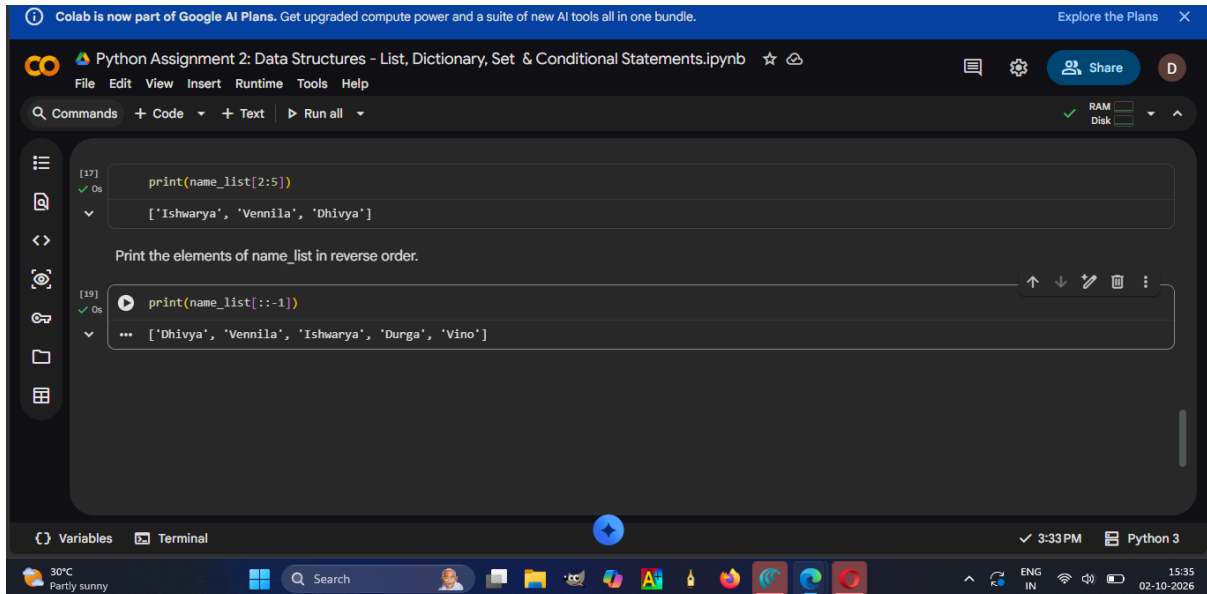
Explanation :

- I had used the slicing method for printing the 2 to 4th element in name_list.
- Here 2 represents the third element 5 represents the last element in the name_list.
- The above mentioned screenshot represents the result.

Task 4: Print the elements of name_list in reverse order.

Solution :

Preview screenshot :



The screenshot shows a Google Colab notebook titled "Python Assignment 2: Data Structures - List, Dictionary, Set & Conditional Statements.ipynb". The notebook has two code cells. The first cell, at index [17], contains the code `print(name_list[2:5])` and its output is `['Ishwarya', 'Vennila', 'Dhivya']`. The second cell, at index [19], contains the code `print(name_list[::-1])` and its output is `['Dhivya', 'Vennila', 'Ishwarya', 'Durga', 'Vino']`. The text "Print the elements of name_list in reverse order." is written above the second code cell. The notebook interface includes a menu bar (File, Edit, View, Insert, Runtime, Tools, Help), a toolbar with icons for file operations and execution, and a status bar at the bottom showing the time as 3:33 PM and the Python version as Python 3. The Windows taskbar is visible at the very bottom of the image.

Explanation :

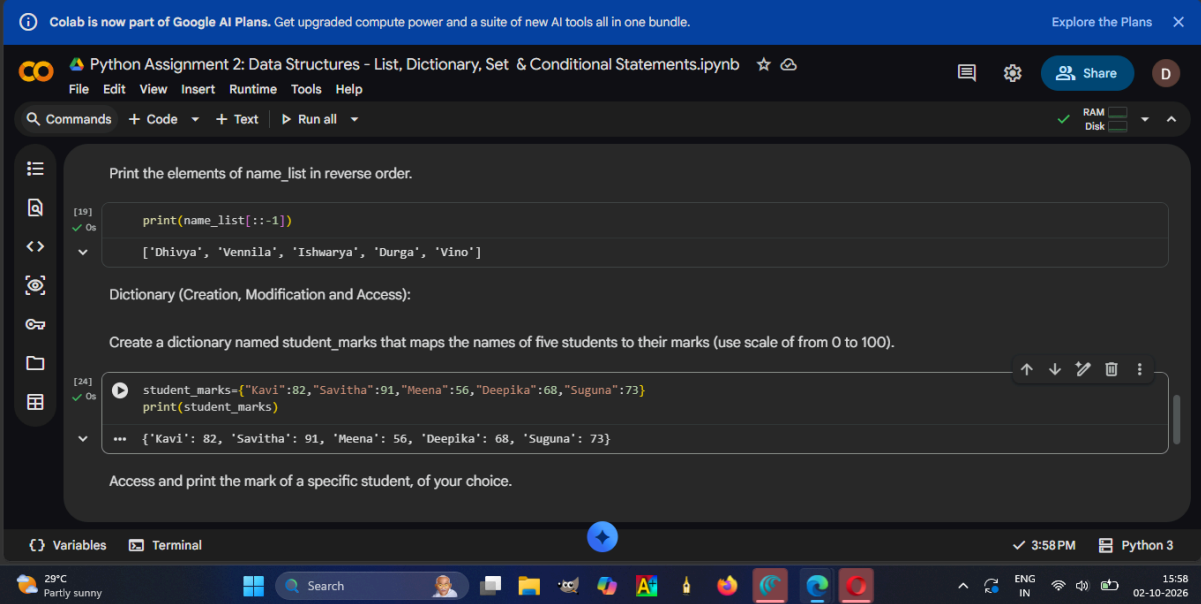
- I had used the slicing method for printing the name_list in reverse order.
- Here `::-1` is used to reverse the name_list.
- The above mentioned screenshot represents the result.

Dictionary (Creation, Modification and Access):

Task 1 : Create a dictionary named `student_marks` that maps the names of five students to their marks (use scale of from 0 to 100).

Solution :

Preview screenshot :



The screenshot shows a Google Colab notebook titled "Python Assignment 2: Data Structures - List, Dictionary, Set & Conditional Statements.ipynb". The notebook contains three code cells. The first cell prints the elements of a list in reverse order. The second cell creates a dictionary named `student_marks` and prints it. The third cell prints the mark of a specific student.

```
[19] print(name_list[::-1])
[19] ['Dhivya', 'Vennila', 'Ishwarya', 'Durga', 'Vino']

Dictionary (Creation, Modification and Access):

Create a dictionary named student_marks that maps the names of five students to their marks (use scale of from 0 to 100).

[24] student_marks={"Kavi":82,"Savitha":91,"Meena":56,"Deepika":68,"Suguna":73}
[24] print(student_marks)
[24] {'Kavi': 82, 'Savitha': 91, 'Meena': 56, 'Deepika': 68, 'Suguna': 73}

Access and print the mark of a specific student, of your choice.
```

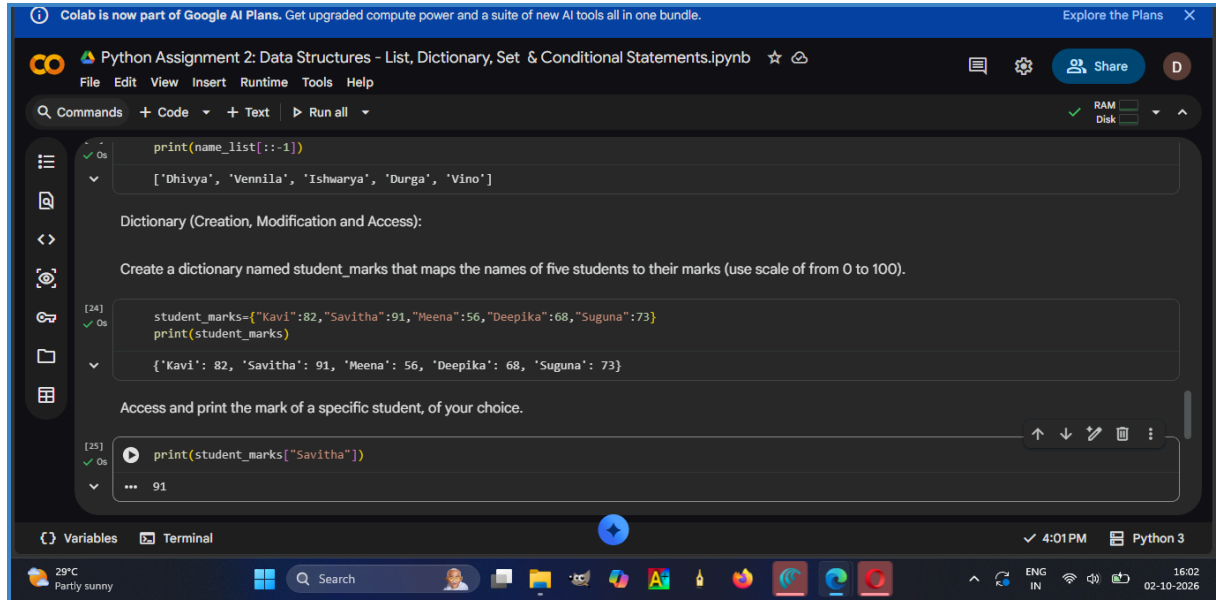
Explanation :

- I had created a dictionary named `student_marks`.
- Here key value pairs are separated by `' : '`.
- The above mentioned screenshot represents the result.

Task 2: Access and print the mark of a specific student, of your choice.

Solution :

Preview screenshot :



The screenshot shows a Google Colab notebook titled "Python Assignment 2: Data Structures - List, Dictionary, Set & Conditional Statements.ipynb". The notebook contains the following code and output:

```
print(name_list[::-1])
```

```
['Dhivya', 'Vennila', 'Ishwarya', 'Durga', 'Vino']
```

Dictionary (Creation, Modification and Access):

Create a dictionary named student_marks that maps the names of five students to their marks (use scale of from 0 to 100).

```
student_marks={"Kavi":82,"Savitha":91,"Meena":56,"Deepika":68,"Suguna":73}
```

```
print(student_marks)
```

```
{'Kavi': 82, 'Savitha': 91, 'Meena': 56, 'Deepika': 68, 'Suguna': 73}
```

Access and print the mark of a specific student, of your choice.

```
print(student_marks["Savitha"])
```

```
91
```

The bottom of the screenshot shows the Windows taskbar with the date 02-10-2026 and time 16:02.

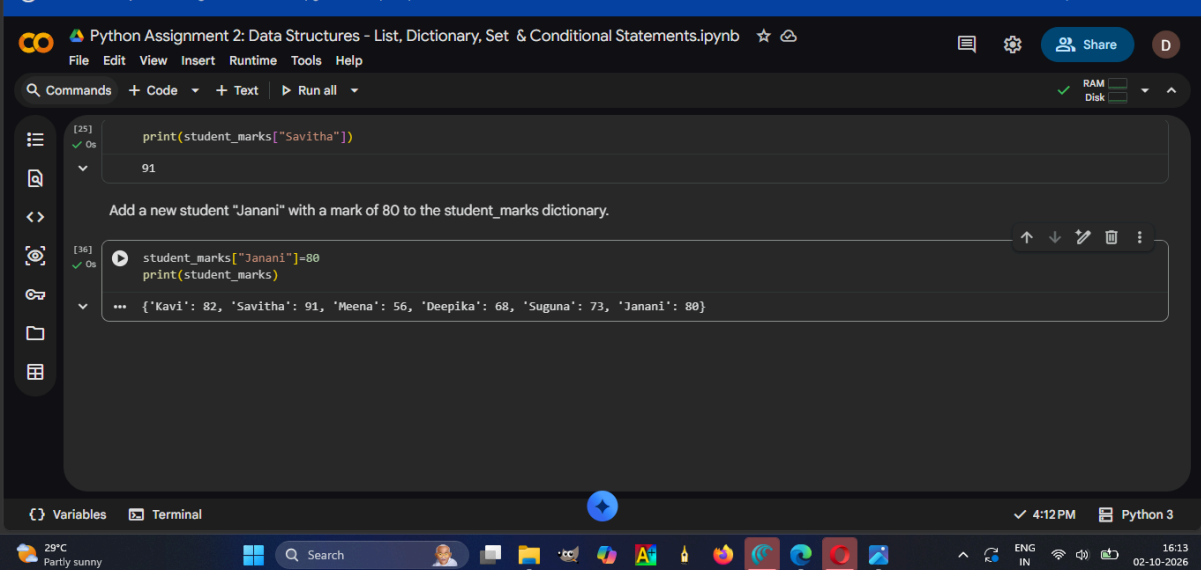
Explanation :

- I had used the key ("Savitha") to print the value (91) in student_marks.
- Here the student name is key and the mark is a value.
- The above mentioned screenshot represents the result.

Task 3 : Add a new student "Janani" with a mark of 80 to the student_marks dictionary.

Solution :

Preview screenshot :



The screenshot shows a Jupyter Notebook interface with the title "Python Assignment 2: Data Structures - List, Dictionary, Set & Conditional Statements.ipynb". The notebook has two code cells. The first cell contains the code `print(student_marks["Savitha"])` and its output is `91`. The second cell contains the code `student_marks["Janani"] = 80` followed by `print(student_marks)`. Below the code, the output of the second cell is displayed as a dictionary: `{'Kavi': 82, 'Savitha': 91, 'Meena': 56, 'Deepika': 68, 'Suguna': 73, 'Janani': 80}`. The notebook interface includes a menu bar with options like File, Edit, View, Insert, Runtime, Tools, and Help. The bottom status bar shows the system time as 4:12 PM and the Python version as Python 3.

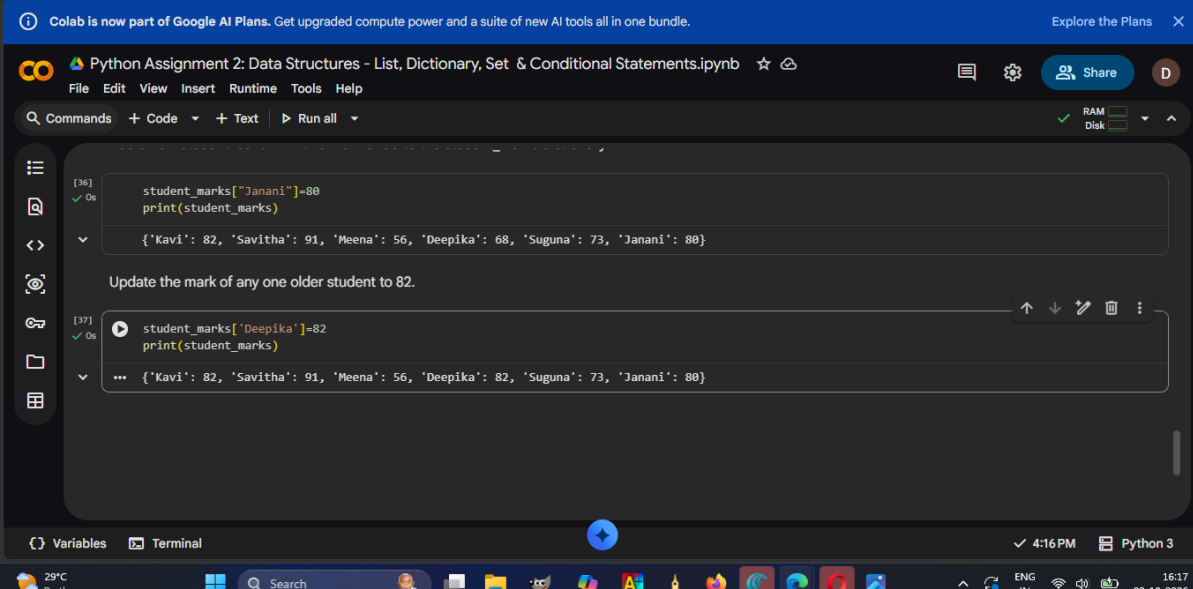
Explanation :

- I had added the student name "Janani" with mark 80 to student_marks.
- The above mentioned screenshot represents the result.

Task 4 : Update the mark of any one older student to 82.

Solution :

Preview screenshot :



The screenshot shows a Google Colab notebook titled "Python Assignment 2: Data Structures - List, Dictionary, Set & Conditional Statements.ipynb". The notebook has two code cells. The first cell, labeled [36], contains the code `student_marks["Janani"]=80` and `print(student_marks)`, which outputs the dictionary `{'Kavi': 82, 'Savitha': 91, 'Meena': 56, 'Deepika': 68, 'Suguna': 73, 'Janani': 80}`. The second cell, labeled [37], contains the text "Update the mark of any one older student to 82." followed by the code `student_marks['Deepika']=82` and `print(student_marks)`, which outputs the updated dictionary `{'Kavi': 82, 'Savitha': 91, 'Meena': 56, 'Deepika': 82, 'Suguna': 73, 'Janani': 80}`. The notebook interface includes a menu bar (File, Edit, View, Insert, Runtime, Tools, Help), a toolbar with icons for commands, code, text, and running all cells, and a status bar at the bottom showing the current time as 4:16 PM and the Python version as Python 3. The Windows taskbar is visible at the very bottom of the image.

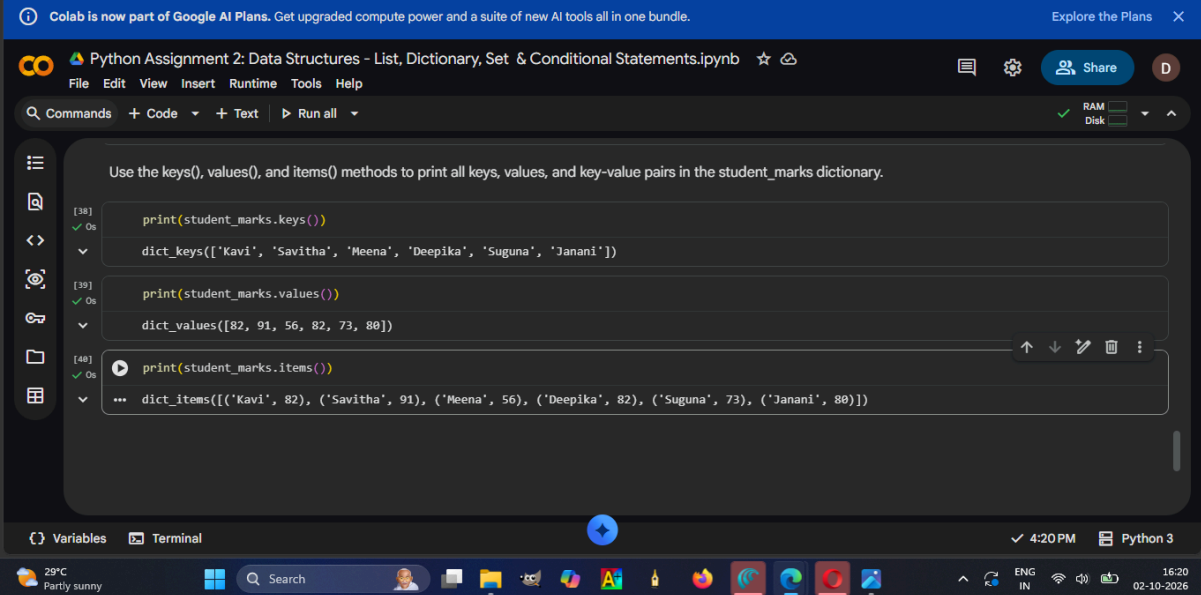
Explanation :

- I had updated the mark 82 to an old student name “Deepika” .
- The above mentioned screenshot represents the result.

Task 5: Use the `keys()`, `values()`, and `items()` methods to print all keys, values, and key-value pairs in the `student_marks` dictionary.

Solution :

Preview screenshot :



The screenshot shows a Google Colab notebook titled "Python Assignment 2: Data Structures - List, Dictionary, Set & Conditional Statements.ipynb". The notebook contains three code cells, each with a comment and a corresponding Python command:

- Cell 1: Comment "Use the keys(), values(), and items() methods to print all keys, values, and key-value pairs in the student_marks dictionary." followed by `print(student_marks.keys())`. The output is `dict_keys(['Kavi', 'Savitha', 'Meena', 'Deepika', 'Suguna', 'Janani'])`.
- Cell 2: `print(student_marks.values())`. The output is `dict_values([82, 91, 56, 82, 73, 80])`.
- Cell 3: `print(student_marks.items())`. The output is `dict_items([('Kavi', 82), ('Savitha', 91), ('Meena', 56), ('Deepika', 82), ('Suguna', 73), ('Janani', 80)])`.

The bottom of the notebook shows a status bar with "Variables", "Terminal", "4:20 PM", and "Python 3". The Windows taskbar is visible at the very bottom.

Explanation :

- I had used the `key` method to print only keys in the student mark dictionary .
- And I had used the `values` method to print only values in the student mark dictionary.
- And then I used the `item` method to print key- value pairs in the student mark dictionary.
- The above mentioned screenshot represents the result.

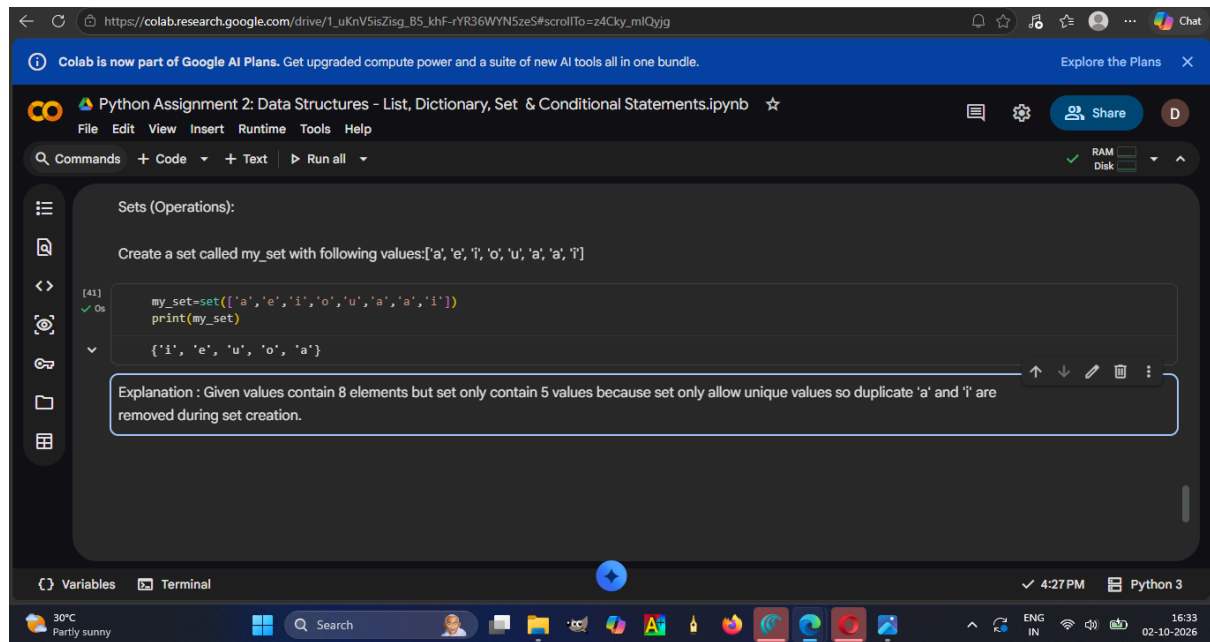
Sets (Operations):

Task 1 : Create a set called `my_set` with following values:

`['a','e','i','o','u','a','a','i']` . Analyse the output and provide explanation for the same.

Solution :

Preview screenshot :



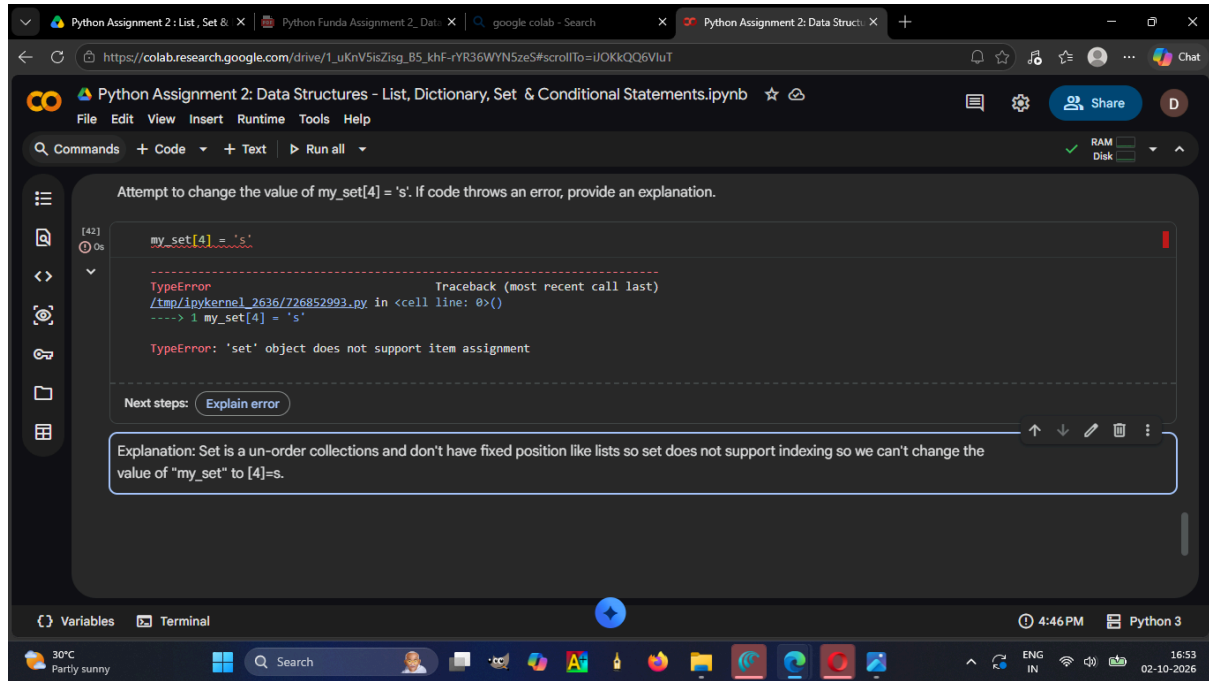
Explanation :

- I had created the set with a given value .
- Given values contain 8 elements but created sets only contain 5 values because sets only allow unique values so duplicates 'a' and 'i' are removed during set creation. .
- The above mentioned screenshot represents the result.

Task 2 : Attempt to change the value of `my_set[4] = 's'`. If code throws an error, provide an explanation.

Solution :

Preview screenshot :



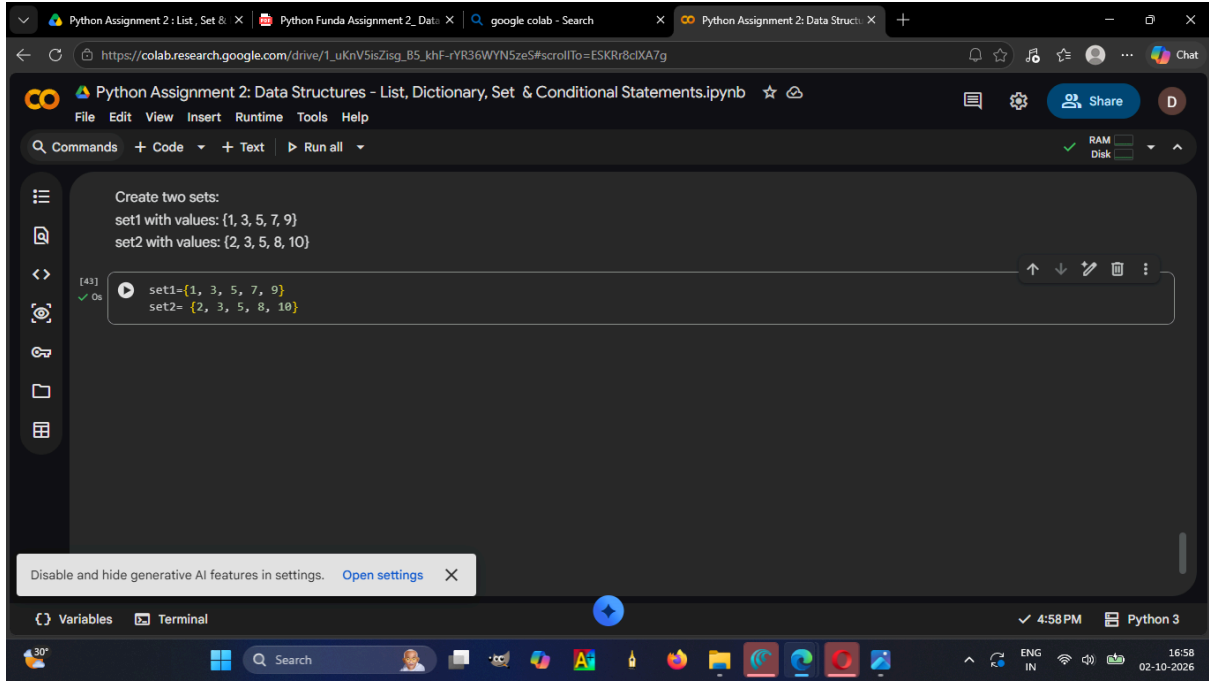
Explanation :

- I had attempted to change the value of `my_set[4]` but it showed an error because set is an un-order collection and doesn't have fixed position like lists so the set does not support indexing so we can't change the value of "my_set" to `[4]=s`.
- The above mentioned screenshot represents the result.

Task 3 : Create two sets: set1 with values: {1, 3, 5, 7, 9} , set2 with values: {2, 3, 5, 8, 10}

Solution:

Preview screenshot :



The screenshot shows a Google Colab notebook interface. The notebook title is "Python Assignment 2: Data Structures - List, Dictionary, Set & Conditional Statements.ipynb". The code cell contains the following text:

```
Create two sets:  
set1 with values: {1, 3, 5, 7, 9}  
set2 with values: {2, 3, 5, 8, 10}
```

Below the text, the code is executed, and the output is displayed:

```
[43]  
✓ 0s  
set1={1, 3, 5, 7, 9}  
set2= {2, 3, 5, 8, 10}
```

The bottom of the screenshot shows the Windows taskbar with the time 4:58 PM and the date 02-10-2026.

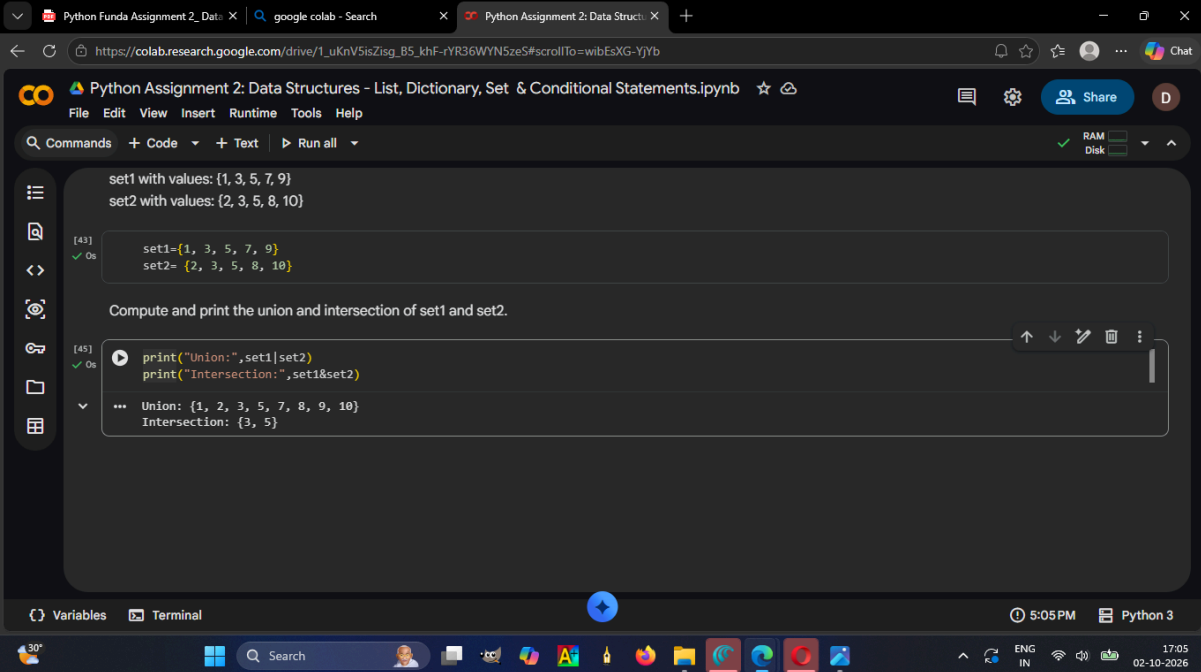
Explanation :

- I had created the two sets named set1 and set2 with the given values.
- The above mentioned screenshot represents the result.

Task 4 : Compute and print the union and intersection of set1 and set2.

Solution:

Preview screenshot :



The screenshot shows a Google Colab notebook titled "Python Assignment 2: Data Structures - List, Dictionary, Set & Conditional Statements.ipynb". The notebook contains the following code and output:

```
set1 with values: {1, 3, 5, 7, 9}
set2 with values: {2, 3, 5, 8, 10}
```

```
[43] ✓ Os
set1={1, 3, 5, 7, 9}
set2={2, 3, 5, 8, 10}
```

Compute and print the union and intersection of set1 and set2.

```
[45] ✓ Os
print("Union:",set1|set2)
print("Intersection:",set1&set2)
```

```
... Union: {1, 2, 3, 5, 7, 8, 9, 10}
Intersection: {3, 5}
```

The bottom of the screenshot shows the Windows taskbar with the time 5:05 PM and date 02-10-2026.

Explanation :

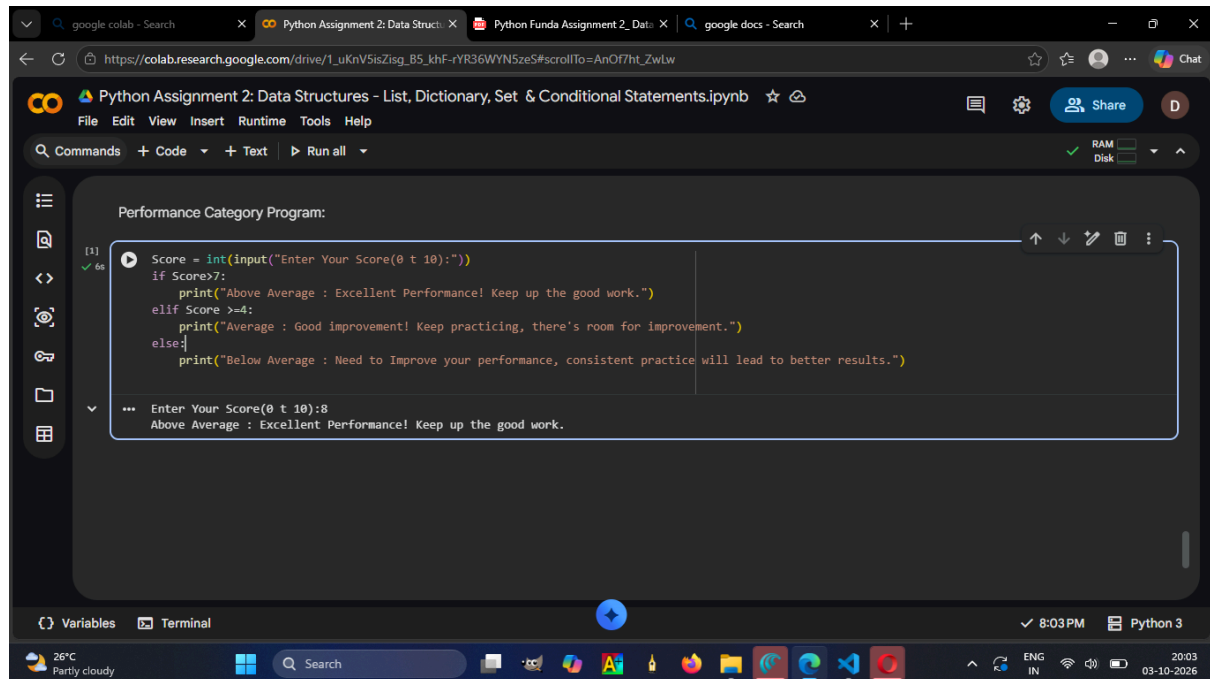
- I had used the “|” symbol for performing union and “&” symbol for performing intersection with set1 and set2 .
- The above mentioned screenshot represents the result.

Operators & Conditional Statements : (IF, ELIF, ELSE)

Task :Write Performance Category Program:

Solution :

Preview screenshot :



The screenshot shows a Google Colab notebook titled "Python Assignment 2: Data Structures - List, Dictionary, Set & Conditional Statements.ipynb". The code in the notebook is as follows:

```
Performance Category Program:

[1] Score = int(input("Enter Your Score(0 t 10):"))
    if Score>7:
        print("Above Average : Excellent Performance! Keep up the good work.")
    elif Score >=4:
        print("Average : Good improvement! Keep practicing, there's room for improvement.")
    else:
        print("Below Average : Need to Improve your performance, consistent practice will lead to better results.")

... Enter Your Score(0 t 10):8
Above Average : Excellent Performance! Keep up the good work.
```

The output of the program is "Above Average : Excellent Performance! Keep up the good work." The notebook interface includes a left sidebar with icons for file management, a top menu bar with options like File, Edit, View, Insert, Runtime, Tools, and Help, and a bottom status bar showing system information like temperature and time.

Explanation :

- I had created the nested conditional statement for the performance category program.
- First it will ask the input score with a message that "The score range should be from 0 to 10 : " .
- Then it will run the if and compare the input with a condition greater than 7 .
- If the score is greater then 7 the print the result as "Above Average : Excellent Performance! Keep up the good work."
- If the score is not greater than 7, and it is greater than or equal to 4 ,it prints the result as "Average : Good improvement! Keep practicing, there's room for improvement."
- If the score is less than 4 then it will print the result as "Below Average : Need to Improve your performance, consistent practice will lead to better results."
- The above mentioned screenshot represents the result.

